

$$\max z = -2x_1 - 3x_2$$

$$\text{s.t. } 2x_1 + 2x_2 \leq 30$$

$$x_1 + 2x_2 \geq 10$$

$$x_1, x_2 \geq 0$$

Solve this problem with dual simplex.

$\leq$ (-).

$$\max z = -2x_1 - 3x_2$$

$$\text{s.t. } 2x_1 + 2x_2 \leq 30$$

$$-x_1 - 2x_2 \leq -10$$

$$x_1, x_2 \geq 0$$

$$\max z = -2x_1 - 3x_2$$

$$2x_1 + 2x_2 + s_1 = 30$$

$$-x_1 - 2x_2 + s_2 = -10$$



E.V. $x_1, x_2, s_1, s_2 \geq 0$.

Basic	x_1	x_2	s_1	s_2	RHS
$\max z \geq 0$	2	$3 \rightarrow 0$	0	0	0
s_1	2	$2 \rightarrow 0$	1	0	30
s_2	-1	-2	0	1	-10

Min Ratio. $\left| \frac{2}{-1} \right|$ $\left| \frac{3}{-2} \right|$ — —

✓ infeasible
✓ opt
(better than opt)

L.V.

	x_1	x_2	s_1	s_2	RHS.	
z	$1/2$	0	0	$3/2$	-15	
s_1	1	0	1	1	20	✓ Feasible
x_2	$1/2$	1	0	$-1/2$	5	✓ Opt
	2	3	0	0	0	
	$-3/2$	-3	0	$3/2$	-15	

STOP

$$\max z = -2x_1 + 3x_2$$

$$\begin{aligned} \text{s.t. } 2x_1 + 2x_2 &\leq 30 && +s_1 \\ x_1 + 2x_2 &\geq 10 && \Rightarrow \leq -10. \\ x_1, x_2 &\geq 0 && +s_2 \end{aligned}$$

Basic	x_1	x_2	s_1	s_2	RHS.
$\max z \geq 0$	2	-3	0	0	0
s_1	2	2	1	0	30
s_2	-1	-2	0	1	-10

✓ infeasible.

✗ Opt. $\Rightarrow$ you cannot apply dual simplex.